

RESEARCH ARTICLE

FinSentiment: Predicting Financial Sentiment Through Transfer Learning

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ABSTRACT

There is an increasing interest in financial text mining tasks. Significant progress has been made by using deep learning-based models on a generic corpus, which also shows reasonable results on financial text mining tasks such as financial sentiment analysis. However, financial sentiment analysis is still demanding work because of the insufficiency of labeled data for the financial domain and its specialized language. General-purpose deep learning methods are not as effective mainly due to specialized language used in the financial context. In this study, we focus on enhancing the performance of financial text mining tasks by improving the existing pretrained language models via NLP transfer learning. Pretrained language models demand a small quantity of labeled samples, and they could be enhanced to a greater extent by training them on domain-specific corpora instead. We propose an enhanced model FinSentiment, which incorporates enhanced versions of a number of recently proposed pretrained models, such as BERT, XLNet, RoBERTa, GPT, Llama, and T5, to better perform across NLP tasks in financial domain by training these models on financial domain corpora. The corresponding finance-specific models in FinSentiment are called Fin-BERT, Fin-XLNet, Fin-RoBERTa, Fin-GPT, Fin-Llama, and Fin-T5, respectively. We also propose variants of these models jointly trained over financial domain and general corpora. Our finance-specific FinSentiment models, in general, show the best performance across three financial sentiment analysis datasets, even when only a subpart of these models is fine-tuned with a smaller training set. Our results exhibit enhancement for each tested performance criteria on the existing results for these datasets. Extensive experimental results demonstrate the effectiveness and robustness of especially RoBERTa pretrained on financial corpora. Overall, we show that NLP transfer learning techniques are favorable solutions to financial sentiment analysis tasks. Our source code has been deposited at <https://github.com/seferlab/finsentiment>.

1 | Introduction

Asset prices in an open economy mirror all the existing knowledge about these assets trading in a market (Malkiel 2003; Gezici and Sefer 2024; Seyhan and Sefer 2023; Tuncer et al. 2022). When novel knowledge is introduced, all performers in the market improve their positions, and accordingly, asset prices are updated. Such adjustments make it almost impossible to consistently outperform the market. On the other hand, “novel knowledge” is

not clearly defined, and its definition may update when novel information retrieval technologies appear. As a result, in the shorter term, early adopters of those technologies may have the edge on many financial tasks such as asset price prediction.

The number of natural language processing (NLP) resources and methods has recently been increasing, especially for more specialized domains (Sefer 2022a; Patro, Duggal, et al. 2012). They have become much more mature and have remarkably

changed the financial domain's landscape. Especially, portfolio managers have a major interest in utilizing NLP methods to track real-time market sentiment from Twitter posts and online news articles, as they can use the extracted sentiment signal for trading assets. Instinctively, a particular company's stock price is expected to drop if negative knowledge exists about the company, and vice versa. For instance, according to Bloomberg, the historical performance of sentiment portfolios is better than the benchmark portfolios by a significant margin (González-Sánchez and Morales Vega 2021). Additionally, prior research on finance finds that a company's performance and market returns can be predicted reasonably well by using the sentiment signal obtained from social media datasets (Tetlock et al. 2008; Sefer and Kingsford 2011).

The volume of financial communication text continues to keep increasing since an incredible number of such texts are being created on a daily basis. In the finance sector, along with various fields, automatically harnessing financial text data is an inevitable measure that should be taken to predict and analyze future market trends in finance and economics. Here, financial text mining partakes an important role in aggregating official company announcements or analyst reports since manual analysis of such texts and gaining actionable insights from them are quite difficult tasks.

Financial sentiment analysis is the main focus of this paper, which categorizes the text as neutral, negative, or positive in the financial domain. We need to handle two challenges while working on this financial sentiment analysis task: (1) The up-to-date sentiment categorization techniques mainly use neural networks while requiring a significant amount of tagged financial datasets. In this case, due to the required background information in the economics or finance fields, annotators for tagging should be the people who are specialized in such fields which restricts the amount of possible annotators. Labeling these finance-specific textual fragments requires experience and expertise which can be extremely expensive. (2) The language of financial statements is task-specific where the statements use a specialized words and they are mainly made up of unclear expressions instead of more easily-comprehended positive or negative words. So, the models trained over a more common corpus do not fit well specifically to the financial sentiment analysis tasks. One way to tackle having less annotated data in finance is to construct language models (LMs), which is a key step to achieve state-of-the-art results in many different NLP tasks, for instance, sentiment analysis which is the main focus of this paper. LMs are techniques used to learn the probability of a given sequence of words occurring in a sentence. There are different kinds of LMs using statistical techniques or neural network algorithms to construct a base for the word predictions. Considering statistical LMs with thoroughly and manually crafted financial sentiment lexicons (Loughran and McDonald 2016) as an answer to the problem of finance having domain-specific language would not be a good solution since such methods are based on word counting, which fails to analyze the deeper semantic meaning of the provided text.

NLP transfer learning approaches come into the minds as an effective strategy to solve the two aforementioned challenges

(Q. Yang et al. 2020). In this case, one can use word embeddings and distributed word representations like GloVe (Pennington et al. 2014) and Word2vec (Mikolov et al. 2013) which are pretrained with a large unlabelled data to initialize the first layer of a neural network model. After that, the remaining layers can usually be trained using a dataset tailored to the specific task (Zaib et al. 2020). While these approaches can grasp the semantic meanings of words, they struggle to identify more complex contextual elements, such as dis-inguishing between multiple meanings of a word, understanding syntactic structures, recognizing semantic roles, and resolving references in discourse (Qiu et al. 2020). The up-to-date methods utilize pretrained LMs to overcome the unsuccessful behavior of context understanding for previous methods. The main point of NLP transfer learning approaches can be summarized in two steps: (1) We train these LMs on a major-size corpus. (2) Then, we initialize the downstream models by using the parameters learned over LM task. By following these two consecutive steps, we can achieve a significantly better performance. In this case, initialized layers may be the overall model (Peters et al. 2018) or even the exclusive word embedding layer (Howard and Ruder 2018). In principle, NLP transfer learning shall become a solution to the problem of scarce tagged data. These transfer learning-based LMs do not ask for associated tags or labels as they are rather based on forecasting the following word. These LMs focus on learning the representation of semantic knowledge. Once such representation is learned, finetuning on the tagged dataset turns into a way to learn the usage of semantic knowledge in predicting the labels.

Transfer learning's main advantage is its capability to additionally pretrain the LMs on domain-specific unlabeled corpora. As a result, the trained model may capture the semantic relationships in the target domain's textual content. This extra semantic relationship learning is important since the distribution of words in the target corpora is probably unlike general corpora. This proposed pretraining method is particularly important and effective for the niche financial domain as its vocabulary and the language content are remarkably dissimilar to a general vocabulary and language. In this research, we investigate three transfer learning-based deep learning models for financial sentiment analysis tasks. One of them, BERT (Bidirectional Encoder Representation from Transformer), (Devlin et al. 2019) is based on Transformer structure (Vaswani et al. 2017; Soylu and Sefer 2024; Sefer 2022b, 2021; Sefer and Kingsford 2021) which is developed around the self-attention concept. Additionally, RoBERTa (Robustly optimized BERT) (Y. Liu et al. 2019) is an optimized version of BERT. XLNet (Transformer-XL Network) (Z. Yang et al. 2019) is another transformer-based model using autoregression. Among more recent large language models (LLMs), Fin-GPT is a finetuned version of the famous GPT model (Brown et al. 2020). Fin-GPT utilizes the lightweight Low-rank Adaptation (LoRA) (Hu et al. 2021) technique to fine-tune open-source LLMs using approximately 50k samples. Fin-Llama is finetuned version of Llama (Touvron et al. 2023) which has fewer parameters compared to Fin-GPT. Similar to Fin-GPT, LoRA is used to fine-tune open-source LLM. Lastly, Fin-T5 is finetuned version of T5 (Text-to-Text Transfer Transformer) LLM (Raffel et al. 2020) for financial tasks.

We have also investigated several features of the introduced models, such as the effects of training strategies in preventing catastrophic forgetting, the effects of extra pretraining over financial corpus, and finetuning solely a tiny subset of model layers in order to decrease the training time without a remarkable performance decrease. Among them, catastrophic forgetting (McCloskey and Cohen 1989) is a common problem in transfer learning, which means the pre-trained knowledge can be removed during learning of new knowledge. Therefore, we also investigate whether various transformer-based models (BERT, RoBERTa, XLNet, Fin-GPT, Llama, and T5) suffer from the catastrophic forgetting problem.

As a summary, we have the following main contributions in this paper:

- We introduce financial extensions of the recently introduced transformer-based models (BERT, RoBERTa, XLNet, GPT, Llama, and T5) for financial NLP tasks, in order to transfer knowledge from financial domain corpora. By calling these financial extensions FinSentiment, we evaluate the performance of FinSentiment over 3 financial sentiment analysis datasets.
- We conduct a number of experiments on several Financial PhraseBank, AnalystTone, and FiQA sentiment scoring benchmark datasets. We achieve state-of-the-art results on all these financial datasets.
- We compare the performance of various transformer-based models (BERT, RoBERTa, XLNet, GPT, Llama, and T5) and understand why certain models can perform better on financial text mining tasks. Our models are capable of efficiently capturing language knowledge and semantic information in large-scale pretraining corpora.
- We have investigated several features of the introduced models, such as the effects of training strategies in preventing catastrophic forgetting, the effects of extra pretraining over financial corpus, and finetuning solely a tiny subset of model layers in order to decrease the training time without a remarkable performance decrease.
- We implement our transformer-based algorithms by using TensorFlow and Hugging Face frameworks (Abadi et al. 2015; Wolf et al. 2019). We make the source code and pretrained models publicly available with a minimal number of task-specific architecture modifications.

The remaining parts of this paper are organized as follows: Section 2 discusses the related work on both more general sentiment analysis as well as finance-specific sentiment analysis via pretrained models. Next, Section 3 defines the evaluated models, including the existing models and FinSentiment. Section 4 follows this section focusing on our experimental setup to evaluate the performance of FinSentiment over the sentiment analysis tasks. Results in Section 5 comes after Section 4 focusing on experimental results on three finance-specific sentiment analysis tasks as well as discussion and analysis of FinSentiment results across multiple points of view. Lastly, Section 6 summarizes our findings and concludes the paper.

2 | Related Work

2.1 | Sentiment Analysis

Sentiment analysis, which is a demanding NLP task for domain-specific fields, infers people's opinions or sentiments from their textual language (Liu 2012). The recent studies can be examined across two different categories: (1) Traditional machine learning approaches where attributes are inferred from the text by word counting (Agarwal and Mittal 2016; Martineau and Finin 2009; Tripathy et al. 2016; Whitelaw et al. 2005) and (2) deep learning-based approaches where a number of embeddings represent a given text (Araque et al. 2017; Severyn and Moschitti 2015; Zhang et al. 2018). More traditional machine learning-based approaches cannot easily represent the semantic knowledge resulting from a particular word sequence. On the other hand, deep learning-based approaches may require a huge amount of data since their parameter space is much larger than the more traditional approaches (Schmidhuber 2015).

Financial sentiment analysis is different than the more general analysis of sentiment in terms of both aim and domain perspectives. Financial sentiment analysis, as a purpose, mainly guesses the reaction of markets to the knowledge introduced in the text (Li et al. 2014). Loughran and McDonald (2016) introduce a detailed analysis on machine learning-based financial text analysis that includes lexicon-based and bag-of-words based approaches. For instance, Loughran and McDonald (2016) generate a financial terms dictionary, where they assign negative, positive, and uncertain tags to the terms in the dictionary and assess a document's tone over extracted words appearing in this dictionary (Loughran and McDonald 2011). Malo et al. (2014) investigate the semantic orientations on economic texts where they have also generated directional words lexicon and financial phrase-bank for financial sentiment analysis. Pagolu et al. (2016) focus on detecting the sentiment by extracting n-grams over tweets having financial knowledge, and feeding them into a machine learning method.

In terms of deep learning approaches, Kraus and Feuerriegel (2017) have applied LSTM (long short-term memory) to analyze financial textual polarity. They have used LSTM over company disclosures in predicting stock price changes, where their approach has outperformed traditional ML-based methods. In comparison to our work, they have pretrained their method over greater corpora. Their method is more restrictive since they pretrain over a labeled dataset whereas our approach FinSentiment pretrains an LM in an unsupervised way. A number of other studies employing neural models also focus on analyzing the financial sentiment. Sohngir et al. (2018) have applied a number of neural network models over a StockTwits dataset and have found CNN (convolutional neural network) as the top performer. Lutz et al. (2019) generate sentence embeddings by Doc2vec over company announcements and predict the stock market by multi-instance learning. Maia, Freitas, et al. (2018) focus on classifying a list of financial text sentences by their sentiment via utilizing LSTM and text simplification. Moreover, aspect-based sentiment analysis (ABSA), as a text analysis type, clusters opinions via aspect and captures the sentiment relevant to each aspect (S. Yang et al. 2018) where deep learning applications have recently started to appear.

Neural networks do not achieve a higher performance in analyzing financial sentiments since labeled financial datasets are missing. Even though the pretrained values are used on the initial neural network layers, the remaining layers yet learn complex associations by using labeled datasets with fewer samples. In this case, a better approach would be to initialize the whole model with pretrained values and fine-tune these values in regard to the following classification task.

2.2 | Text Classification Over Pretrained Models

Language modeling predicts the succeeding word for a given text. Recently, it has been realized that we can favorably apply a NLP model trained for language modeling to the remaining downstream NLP tasks with minor changes by finetuning. Those models are in general trained over a larger corpus, and then are finetuned over the target dataset by integrating extra task-specific layers (Kant et al. 2018). One main example use case for such method is text classification, as focused in this paper. ELMo (Embeddings from LMs) (Peters et al. 2018) can be considered as the initial application of such finetuning task. ELMo includes a bidirectional deep LM which is pretrained on a larger corpus. It comes up with a contextualized representation for each word by using the hidden states. Using the contextual embeddings for downstream tasks enhances the performance over many tasks when compared to using the static embeddings extracted by Word2vec (Mikolov et al. 2013). Another model ULMFit (Universal Language Model Fine-tuning) (Howard and Ruder 2018) achieves transfer learning for NLP tasks by integrating novel approaches such as gradual unfreezing, slanted triangular learning rates, and discriminative finetuning which sustain information during the training period. They have additionally pretrained LM on domain-specific corpora, where they assumed that the distribution of the target tasks dataset is not the same as the one which is used to train the initial model.

Transformer-based LMs are utilized in several different fields of NLP (Min et al. 2021; Patro, Sefer, et al. 2011). As an example, BERT (Bidirectional Encoder Representations from Transformers) (Devlin et al. 2019) has enhanced this principal idea of effectively finetuning a pretrained LM for downstream tasks. BERT has two key differences from the previous work: (1) BERT is composed of a huge neural network which is trained on an extremely large corpus, and (2) BERT focuses on forecasting randomly-masked tokens over a given token sequence instead of the subsequent token, as well as on classifying two sentences whether they follow each other or not. Due to those two enhancements, BERT has achieved the best results in a number of NLP tasks including question answering. Z. Yang et al. (2019) have introduced a generalized autoregressive pretraining approach XLNet which has two key differences than BERT: (1) XLNet's autoregressive formula is useful in solving a number of limitations of BERT, (2) XLNet learns bidirectional contexts via optimizing the expected likelihood over the factorization order's all possible permutations. In comparison to XLNet, BERT depends on noising the input sequence with masks. It suffers from a pretrain finetune discrepancy by neglecting the dependencies among the masked parts. XLNet has significantly outperformed BERT on a number of tasks such as sentiment analysis, document ranking, etc. Similarly, Y. Liu et al. (2019) have found that

BERT has been remarkably undertrained, and as a result, introduced RoBERTa which is an enhanced version of BERT in terms of training size and multiple principal hyperparameters. Their results exhibit the importance of previously overlooked design criteria in BERT. On the other hand, GPT v3 is also a transformer-based autoregressive LLM approach performing quite well in various general text benchmarks. GPT is developed to scale up LMs which significantly enhances the performance. Llama is a collection of foundation LMs ranging from 7B to 65B parameters, where they trained the models without using private datasets which cannot be easily accessed. Finally, T5 is a unified Text-to-Text Transformer. It follows an encoder-decoder structure and is pretrained on a diverse set of both supervised and unsupervised tasks, all reformulated into a text-to-text format. By finetuning all these models via financial data, we are not interested in comparing them directly, but we also want the impact of transfer learning on financial domain. The details of finetuning for these recently introduced models BERT, XLNet, RoBERTa, GPT, Llama, and T5 have not been analyzed in detail for text classification. One such study (Sun et al. 2019) has carried out experiments over a number of BERT configurations for text classification.

There are a number of previous attempts to use BERT model on financial corpora (Araci 2019; Z. Liu et al. 2020; Q. Yang et al. 2020). Z. Liu et al. (2020) have proposed a model named FinBERT where they have replaced the masked language modeling and next sentence prediction stages of BERT with various different kinds of financial text mining tasks such as dialogue relation task, sentence distance task, and token-passage prediction task. They have outperformed BERT across various domain-specific tasks, where they have also trained FinBERT on both general and financial corpora. Additionally, Araci (2019) has also come up with a different FinBERT model to investigate the prediction performance for the same tasks. He pretrained a BERT model once more with a financial dataset to compare it with the previously suggested ULMFit and ELMo models, as well as to arrange an experimental setup in a way that catastrophic forgetting effects and layer contributions on FinBERT have been observed. Furthermore, Q. Yang et al. (2020) have introduced another FinBERT-named model. They have finetuned BERT over financial datasets including analyst reports in accounting and finance, where their study has indicated another success of domain-specific pretraining over the general BERT model. In addition to all, one more FinBERT model has been introduced by Desola et al. (2019), which uses SEC filings for a number of NLP tasks such as next sentence prediction and masked language modeling. They have also investigated the language change for finance, and their models outperformed BERT on these NLP tasks. In other attempts, Yang et al. (2023) and Todt et al. (2023) have been proposed. However, they are not trained on a variety of financial tasks as we did.

Furthermore, solutions to domain-specificity have been introduced in different research areas, such as defining a specialized language for biomedical studies. Lee et al. (2020) have implemented BioBERT for biological text mining tasks, which has shown outstanding performance over a number of such tasks including biomedical question answering, biomedical relation extraction, and biomedical named-entity recognition. Asgari and Mofrad (2015) have presented an unsupervised data-driven

approach to represent and extract features for biological sequences, which are in turn used in a number of machine learning applications.

3 | Methods

In recent years, unsupervised pretraining of LMs on large text corpora has greatly enhanced the performance of various NLP tasks. These models are typically trained on broad datasets, such as Wikipedia. However, sentiment analysis is highly dependent on the specific domain. The financial industry has amassed vast amounts of financial and business communication texts. Consequently, combining the strengths of unsupervised pretraining with this extensive financial data has the potential to improve a wide range of financial applications. In this work, we introduce our domain-specific adaptations of BERT, XLNet, and RoBERTa—named Fin-BERT, Fin-XLNet, and Fin-RoBERTa—after first providing background information on the underlying neural network models.

3.1 | Preliminaries

3.1.1 | LSTM

LSTM (Hochreiter and Schmidhuber 1997), as a kind of recurrent neural network (RNN), uses update and forget gates to better model the persistence of long-term dependencies in a given sequential data. LSTM is one of the most common techniques to model a sequential data creation process, ranging from natural language to asset prices. As a text is composed of a token sequence, deciding the initial representation of an individual token is the initial target for any LSTM NLP model. Then, one usual practice is to use pretrained weights for representing the token initially. GloVe (Global Vectors for Word Representation) (Pennington et al. 2014) is one of such pretraining approaches. GloVe model calculates word embeddings by training a regression model over word co-occurrence distribution on larger corpora. Even though GloVe represents words effectively, its embeddings are not contextualized in regards to the sequence they are part of. On the other hand, ELMo (Peters et al. 2018) represents the words as contextualized embeddings, where embedding of a word is influenced via its neighboring words. ELMo mainly uses bidirectional LM having several LSTM layers. LM learns probability distribution on a consecutive series of tokens over the provided vocabulary. ELMo focuses on modeling the probability of a token, given subsequent and preceding tokens in a token sequence. The model calculates a single contextualized embedding for each token, by learning a scheme to weight various embeddings over multiple LSTM layers. Once the model infers the contextualized embeddings, those embeddings may be utilized in initializing the downstream NLP tasks.

3.1.2 | Transformers

Transformer uses an attention-based mechanism to model the knowledge in a sequential data (Vaswani et al. 2017). Transformers are great alternatives to RNNs, and they have

encoder–decoder architecture as a sequence-to-sequence model. The encoder in a transformer includes a number of identical layers. Every single layer consists of a fully connected feedforward neural network and multi-head self-attention layer. Transformer learns three mappings, value, query, and key, from the embeddings over a single self-attention layer. It calculates a similarity score by multiplying every token's key vector with query vectors of all tokens. The model arrives at token's newer embedding via weighting the value vectors according to those similarity scores. In a multihead self-attention mechanism, the model concatenates those layers all together, such that it could evaluate a token sequence according to different perspectives. Following such concatenation, the resulting vectors undergo fully connected layers that share parameters. Here, we have focused on explaining solely the encoder part in more detail. However, the decoder part is extremely alike. Transformer has a number of advantages over recurrent neural network models (Vaswani et al. 2017). For instance, RNNs cannot be easily parallelized over GPUs due to their sequential nature. Additionally, information cannot persist due to excessive number of steps among distant items in a sequence.

3.1.3 | BERT

BERT (Devlin et al. 2019), as a LM, is composed of a number of transformer encoders added over each other. But, its LM tasks are defined different than LSTM and ELMo. BERT does not focus on forecasting the following token given subsequent tokens. Instead, one task it focuses on is to mask a randomly selected 15% of the whole set of tokens. These masked tokens are forecasted by using a softmax layer over the final encoder layer. Predicting the next sentence is another task for BERT training. Given two sentences, BERT forecasts whether these sentences come after each other. It represents each input token sequence with position and token embeddings. [SEP] and [CLS] tokens are appended to the termination and start of the sequence, respectively. It uses [CLS] token during all classification tasks. BERT comes up with two versions: (1) BERT-base version has 12 encoder layers with latent size being 768. It has 12 multihead attention heads and is defined over 110M trainable parameters; (2) BERT-large version has 24 encoder layers with latent size being 1024. It has 16 multihead attention heads and is defined over 340M trainable parameters in total.

3.1.4 | XLNet

XLNet is a generalized auto-regressive LM which is developed considering the vulnerabilities of BERT-type models caused from degradation of data based on masking (Z. Yang et al. 2019). The aim is to get rid of such flaws by using probabilities of all permutations in the context. XLNet utilizes a permutation-based objective that combines the autoregression and autoencoding advantages. In summary, XLNet has two main advantages and enhancements: (1) It learns bidirectional context via optimizing the likelihood over the factorization sequence's whole set of permutations. (2) Its autoregression-based formula allows it to overcome BERT's limitations. Additionally, XLNet incorporates the Transformer-XL concepts into pretraining, which is an advanced autoregressive model. Their greatest model XLNet-Large

uses the same architectural parameters as BERT-large, where their model sizes become the same. XLNet is found to achieve remarkable performance enhancement over BERT on multiple NLP tasks.

3.1.5 | RoBERTa

RoBERTa is basically an adjusted version of BERT according to the study of Liu and her colleagues (Y. Liu et al. 2019). They found BERT to be remarkably undertrained and modified the effect of multiple principal hyperparameters. The following levels of the BERT model have been optimized: (1) They implement dynamic masking for the masked language modeling stage, (2) they eliminate the next sentence prediction objective and loss, (3) they train with large mini-batches leading to viable speed and accuracy, and (4) they use Byte-Pair Encoding instead of Unicode Encoding. RoBERTa is found to achieve a performance enhancement over BERT on multiple NLP tasks.

3.1.6 | GPT

GPT is a generalized autoregressive LM which is developed by an independent group other than BERT, XLNet, and RoBERTa. We will use GPT-3 for finetuning to financial text. Overall, GPT-3 uses the identical architecture and model as GPT-2 (Radford et al. 2019) where reversible tokenization, prenormalization, and modified initialization are common in both GPT-2 and GPT-3. The only exception is that GPT-3 uses alternating locally banded sparse attention patterns and is dense in the layers of the transformer which is similar to Sparse Transformer (Child et al. 2019). In their study, the different models of different sizes with orders of magnitude range from 125 million parameters to 175 billion parameters where the last model becomes GPT-3. As part of training, they have also integrated one-shot and few-shot trainings (Radford et al. 2019).

3.1.7 | Llama

LLama is a generalized transformer-based LLM. They integrate multiple enhancements that are used in other models such as PaLM (Chowdhery et al. 2022). Those changes are as follows:

- Pre-normalization is introduced in (Radford et al. 2019). Prenormalization enhances the stability of training where it normalizes each transformer sublayer's input instead of normalizing the output. Normalizing the output is more common, but Llama shows better performance of prenormalization. Prenormalization is based on the RMSNorm normalizing function (Zhang and Sennrich 2019).
- SwiGLU activation function is introduced in PaLM (Chowdhery et al. 2022). They use SwiGLU activation function instead of ReLU. They also use fewer dimensions than PaLM.
- Rotary Embeddings, which are introduced in GPT-Neo (Black et al. 2021), do not use absolute positional embeddings, but they add rotary positional embeddings (Su et al. 2023) instead at the network.

3.1.8 | T5

T5 is a unified framework that transforms all textual problems into a common text-to-text format. By such conversion and pre-training a text-to-text transformer on a larger text corpus, they obtain state-of-the-art performance on a distinct set of NLP tasks. T5 is transformer-based, where it follows the originally proposed encoder-decoder transformer implementation by Vaswani et al. (Vaswani et al. 2017). First of all, they map the sequence of input tokens to a set of embeddings and feed them to the encoder. Each encoder is made up of a set of blocks that have a self-attention layer followed by a small feed-forward network in them. The decoder is quite similar to the encoder structure, but instead, it just includes standard attention. Due to the order-independent feature of self-attention, they use relative positional embeddings which are more than learned positional embeddings. Relative position embeddings do not use a fixed embedding for each position.

3.2 | Fin-BERT: BERT for Financial Domain

Here, we define the enhanced BERT model for the financial domain, where Fin-BERT is finetuned version of BERT. We will focus on three major components: (1) How will additional pretraining on financial domain corpora be achieved by Fin-BERT, (2) How will Fin-BERT be implemented for text regression and classification tasks, and (3) How will Fin-BERT block catastrophic forgetting by integrating additional training strategies as part of finetuning.

3.2.1 | Additional Pretraining

According to Howard and Ruder (2018), the final classification results can be enhanced by additional pretraining of an LM on target domain corpora. Similarly, Araci (2019), Z. Liu et al. (2020), and Q. Yang et al. (2020) also show an enhanced performance for BERT across the financial domain over classification results. We carry out experiments with two separate independent techniques for additional pretraining. The first technique is to pretrain the network on a significantly larger corpus from the target task domain. In this technique, we pretrain a BERT LM over a financial corpus, where corpus details are defined in Section 4.1.1. The second technique is to pretrain the network solely over the training classification dataset sentences. Even though the second corpus is remarkably more compact, utilizing the dataset directly from the target may adapt better to the target domain.

3.2.2 | Fin-BERT for Text Classification and Regression

For sentiment classification, we have added a dense layer just after the final hidden layer's [CLS] token state. This practice is the recommended approach to use BERT in classification tasks (Devlin et al. 2019). Afterwards, we train the classifier neural network over the labeled sentiment dataset. Even though this paper focuses on classification, we have also implemented regression by using an almost similar framework over a distinct dataset having continuous target values. The only principal difference between classification and regression frameworks is that classification uses cross-entropy loss, whereas regression uses mean squared error.

3.2.3 | Preventing Catastrophic Forgetting

This fine-tuning method comes with the major risk of catastrophic forgetting as discussed previously by (Howard and Ruder 2018). This major catastrophic forgetting risk is due to the model rapidly forgetting the knowledge from the LM task when it is being fine-tuned for the new task. We handle this catastrophic forgetting risk by applying three approaches as introduced by (Howard and Ruder 2018): gradual unfreezing, discriminative fine-tuning, and slanted triangular learning rates.

The slanted triangular learning rate approach uses a nonuniform learning rate throughout training. Learning rate schedule in the form of slanted triangle is applied, where learning rate initially climbs up to a certain point and linearly drops afterwards. On the other hand, discriminative fine-tuning uses smaller learning rates for neural network's lower layers and higher rates for network's shallow layers. Let the learning rate at l -th layer be α . Given θ discrimination rate, we use $\alpha_{l-1} = \theta\alpha_l$ as the learning rate for $l-1$ th layer. In discriminative fine-tuning, fine-tuning different layers nonuniformly makes sense since lower layers model the deeper language knowledge whereas the upper layers incorporate the knowledge for the actual classification task. Lastly, gradual freezing initially trains the network over all layers except the classifier layer which is frozen. Then, as part of the training, we slowly unfreeze all layers beginning from the highest layer, such that the least fine-tuned features correspond to lower layer features. As a result, the model is blocked from forgetting lower level language knowledge throughout the initial training stages, which has been learned over pretraining.

3.3 | Fin-XLNet: XLNet for Financial Domain

Similar to the additional pretraining of Fin-BERT in Section 3.2, we carry out experiments with the same 2 techniques for additional pretraining. The first technique is to pretrain the network on a significantly larger corpora from the target task domain, whereas the second technique is to pretrain the network solely over the training classification dataset sentences. Similar to the Fin-BERT, we have added a dense layer just after the final hidden layer's [CLS] token state in terms of sentiment classification. The rest of the process is similar for both classification and regression. In this case, classification again utilizes cross-entropy loss whereas regression utilizes mean squared error. Lastly, Fin-XLNet may also suffer from the major risk of catastrophic forgetting as discussed previously. Similar to Fin-BERT, we handle this catastrophic forgetting risk by applying three approaches as introduced by (Howard and Ruder 2018): gradual unfreezing, discriminative fine-tuning, and slanted triangular learning rates.

3.4 | Fin-RoBERTa: RoBERTa for Financial Domain

Additional pretraining, application of Fin-RoBERTa to classification and regression tasks and incorporating training strategies in Fin-RoBERTa to prevent catastrophic forgetting are similar to Fin-BERT as discussed in Section 3.2.

3.5 | Fin-GPT/Fin-Llama/Fin-T5: GPT/Llama/T5 for Financial Domain

Similar to the approach in Section 3.3, we experimented with the identical two approaches to extra pretrain for Fin-GPT, Fin-Llama, and Fin-T5. The first approach is based on pretraining with larger corpora, whereas the second approach is based on pretraining architecture only over the training classification dataset sentences. In the end, we augmented the architecture with a fully connected layer just after the last hidden state, which is required for sentiment classification. The rest of the process is similar for both classification and regression. In this case, classification again utilizes cross-entropy loss, whereas regression utilizes mean squared error.

4 | Experimental Setup

4.1 | Datasets

4.1.1 | Corporate Filings 10-K and 10-Q

Corporate reports and filings are one of the most crucial textual datasets in business communication and finance. In the United States, all publicly traded companies are required to file annual reports, known as 10-K, and quarterly filings, known as Form 10-Q, mandated by the Securities Exchange Commission (SEC). 10-K and 10-Q filings carry out detailed information on the filing company's financial and business conditions. Companies cannot make misleading or false statements in these filings, as prohibited by regulation and law. Publicly available 10-K and 10-Q filings could be accessed from the SEC website. Here, we obtain 60490 10-K and 142622 10-Q filings of Russell 3000 companies between 1994 and 2019 from the SEC website. We focus on the subset of sections with textual parts in our analysis; Item 1 (Business) in 10-Ks, Item 1A (Risk Factors) in both 10-Ks and 10-Qs, and Item 7 (Managements Discussion and Analysis) in 10-Ks.

4.1.2 | Financial PhraseBank

We mainly focus on Financial PhraseBank dataset for sentiment analysis tasks (Malo et al. 2014) which is publicly available in (Malo et al. 2013). This study utilizes a dataset composed of English language news articles related to all publicly listed companies on the OMX Helsinki exchange. From the full collection, a randomly selected subset of 10,000 articles was chosen to ensure broad representation—covering both large and small firms, various industries, and diverse news sources. In line with the methodology of (Maks and Vossen 2010), all sentences that did not include any entities from the specified lexicon were filtered out. This step narrowed the dataset to 53,400 sentences, each containing at least one identified lexicon entity. The remaining sentences were then categorized based on the types of entity sequences they contained. The final Financial PhraseBank dataset is composed of 4840 English sentences chosen randomly from a financial news database called LexisNexis. Each sentence is then labeled by people with finance or economics background. As part of such annotation, the annotating people have been

requested to assign labels to sentences with respect to the possible effect of the sentence content on the corresponding company's stock price. In addition, the dataset incorporates knowledge about the agreement degrees over the sentences between annotators. These sentences are grouped according to the agreement degree of the annotators, such as 100%, 75%–99%, 66%–74%, 50%–65%. Table 1 summarizes the dataset in terms of sentiment labels and agreement degrees. We utilize 10-fold cross-validation for some of the experiments.

4.1.3 | FiQA Dataset

FiQA dataset (Maia, Handschuh, et al. 2018) contains a total of 1174 examples of news headlines and tweets, and it has specifically been generated for financial question answering and opinion mining challenges. Each example contains the sentence, the sentence snippet associated with the target financial entity, aspects, and sentiment score. A sample FiQA entry is shown in the Table 2. Different than Financial PhraseBank, the sentiment score takes continuous value in the interval of -1 and 1 where 1 represents the most positive case. A Level 1 Aspect label takes on one of 4 possible labels (Corporate, Economy, Market or Stock), and Level 2 Aspect label takes on one of 27 possible labels (Appointment, Risks, Dividend Policy, Financial, Legal, Volatility, Coverage, Price Action, etc.). Tenfold cross-validation is performed for the assessment of the models over this dataset.

4.1.4 | AnalystTone Dataset

AnalystTone dataset (Huang et al. 2014) measures judgments and beliefs in analyst reports that are frequently employed in Finance and Accounting literature. The dataset consists of 10,000 randomly selected sentences over analyst reports in

TABLE 1 | Agreement degrees and sentiment tags on Financial PhraseBank dataset.

Agreement degree	Count	Neutral	Positive	Negative
100%	2259	61.4%	25.2%	13.4%
75%–99%	1189	63.6%	26.6%	9.8%
66%–74%	763	50.9%	36.7%	12.3%
50%–65%	629	54.5%	31.1%	14.4%
All	4840	59.4%	28.1%	12.4%

TABLE 2 | An example entry from FiQA.

Sentence	easyJet expects resilient demand to withstand security fears
Aspect level 1	Corporate
Aspect level 2	Risks
Sentiment score	0.165
Target	easyJet

Investext database. Each sentence is labeled into one of the three groups: neutral, positive, and negative. As a result, the dataset is not balanced as it contains 4590 neutral, 3580 positive, and 1830 negative sentences. Tenfold cross-validation is performed for the assessment of the models over this dataset.

4.2 | Preprocessing

Before feeding 10-K and 10-Q corpus for pretraining, each sentence is separated line by line and an empty line is inserted between each different document. In the sentiment analysis part, for the Financial PhraseBank and AnalystTone datasets, label “positive” is converted to 1 , label “negative” is converted to -1 and label “neutral” is converted to 0 . However, for FiQA datasets, each sentences is labeled according to the corresponding sentiment score. Sentiment scores greater than 0.33 are labeled as 1 , those that are less than -0.33 are labeled as -1 , and those in between are labeled as 0 . We would expect that the model trained with FiQA headline will perform better than other models trained by Financial PhraseBank datasets for predicting FiQA post labels, and vice versa.

4.3 | Baseline Methods

To better evaluate the performance of our domain-specific models Fin-BERT, Fin-XLNet, and Fin-RoBERTa, we consider the following seven baselines where six of them are the original versions of introduced models.

4.3.1 | LSTM Classifier

We implement two bidirectional LSTM classifiers: LSTM classifier with GloVe (Pennington et al. 2014) embeddings and LSTM classifier with ELMo (Peters et al. 2018) embeddings. Both models are the same except one uses ELMo embeddings, whereas the other one uses GloVe embeddings. In both classifiers, we use a hidden size of 128 for all states except the last one where the last hidden state dimension is 256 mainly because of bidirectionality. Then, the models have a fully connected feedforward layer mapping the last latent state to a three-dimensional vector which represents the three labels probabilities. We use a learning rate of $3e^{-5}$ and 0.3 dropout probability across both models. Both models are trained till no improvement is observed for 10 epochs in the validation dataset loss. In the results section, we report the best result out of these two models under the name LSTM.

4.3.2 | BERT

We use the default BERT parameters as discussed in (Devlin et al. 2019). BERT is trained with 256 batch size over a million steps. Adam optimizer is used with a learning rate of $1e^{-4}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay of 0.01 for L_2 . The learning rate decays linearly, and warmup is applied for the first 10,000 steps for learning rate warmup. A 0.1 dropout probability is used for the whole set of layers. GELU activation is used instead of more traditional RELU. BERT training loss is the summation of two

likelihoods: the average masked LM likelihood and the average following sentence forecasting likelihood.

4.3.3 | XLNet

We use the default XLNet parameters as discussed in (Z. Yang et al. 2019). Their largest model XLNet-Large's architectural parameters are the same as BERT-large, resulting in the same model dimensions. As part of the pretraining, the length of the used sequences is 512. Adam optimizer is run for half a million steps where the learning rate decays linearly and batch size is 8192.

4.3.4 | RoBERTa

We use the default RoBERTa parameters as discussed in (Y. Liu et al. 2019). RoBERTa is trained similarly to BERT-large architecture.

4.3.5 | GPT

We use the default largest GPT model's architectural parameters as discussed in (Brown et al. 2020) for this autoregressive model, which has 175 billion parameters, 96 heads in multi-head attention part, and 96 layers in total. Each bottleneck layer uses 12,288 number of units, and the context window is 2048 tokens. The dimension of each multihead attention layer is 128. We use the same batch size of 3.2 million and 0.6×10^{-4} learning rate with Adam optimizer (Kingma and Ba 2017). This GPT-3 model is trained by using 300 billion tokens in total. Overall, they select these parameters in order to minimize data transfer within nodes across GPUs and for computational efficiency purposes.

4.3.6 | Llama

We use the largest Llama model's architectural parameters as discussed in (Touvron et al. 2023) which has 65.2 billion parameters, 64 heads in multihead attention part, and 80 layers in total. The dimension of each multihead attention layer is 8192. We use the same batch size of 4 million and 1.5×10^{-4} learning rate with Adam optimizer (Kingma and Ba 2017). This GPT-3 model is trained by using 1.4 trillion tokens in total. Similar to GPT, they select those parameters in order to minimize data transfer within nodes across GPUs and for computational efficiency purposes.

4.3.7 | T5

We use the default largest T5 model's architectural parameters as discussed in Raffel et al. (2020) where the denoising objective is used. It has 250 billion parameters and 96 heads in the multi-head attention part. The dimension of each multihead attention layer is 128. The learning rate of their Adam optimizer training is 10^{-4} . Overall, they select those parameters in order to minimize data transfer within nodes across GPUs and for computational efficiency purposes.

4.4 | Performance Evaluation

We evaluate the models as part of classification in terms of cross-entropy loss, macro F1 average, and accuracy. Accuracy is calculated as the number of correct predictions over the number of all predictions. Macro F1 average first estimates F1 scores for each class and then averages them out, where $F1 = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$. In this case, precision is the ratio of true positives (TPs) to the predicted positive instances ($\frac{TP}{TP+FP}$), whereas recall is the proportion of TPs to the actual positive instances ($\frac{TP}{TP+FN}$). F1 score is another successful metric to evaluate the classification performance, which is useful for Financial PhraseBank dataset where labels are not quite balanced (approximately 60% of the sentences are neutral). Cross-entropy loss is computed when adjusting model weights, aiming to minimize the loss. Cross-entropy loss is basically a measure of the difference between two probability distributions for a given random variable or set of events. While evaluating the regression model, we use standard metrics such as R^2 and mean squared error. Among these evaluation metrics, higher F1, higher accuracy, and lower loss scores indicate better performance.

4.5 | Implementation Details

In Fin-BERT, we use the following metrics: Trained sequence is maximally made up of 64 tokens, warm-up ratio is 0.2, learning rate becomes $2e^{-5}$, and the dropout probability is 0.1. The models are trained over 15 epochs, and the best models are selected over evaluation results on the validation set. The discrimination ratio is used as 0.85 as part of discriminative finetuning. In the remaining methods (Fin-XLNet, Fin-RoBERTa, Fin-GPT, Fin-Llama, and Fin-T5), we mainly use the default hyperparameters (Y. Liu et al. 2019; Z. Yang et al. 2019). Apart from these parameters, all these models are again trained over 15, and the best models are selected over evaluation results on the validation set. The discrimination ratio is used as 0.85 as part of discriminative finetuning. We have used the transformers from the Hugging Face library (Wolf et al. 2019) for pretraining and fine-tuning purposes. We have extracted the results for pretrained models uploaded in the Hugging Face library. For BERT, BERT-base-cased and BERT-large-cased models are used; for RoBERTa, roberta-base and roberta-large are used; and for XLNet, XLNet-base cased and XLNet-large cased are used.

5 | Results and Discussion

5.1 | Finance-Specific FinSentiment Models Outperform the General Models

Table 3 shows the results of finance-specific FinSentiment and general baseline methods (LSTM, BERT, XLNet, and RoBERTa) that are state-of-the-art for the sentiment classification task in the Financial PhraseBank dataset over 10-fold cross-validation. In this case, our model FinSentiment in the table represents the best performance over Fin-BERT, Fin-XLNet, and Fin-RoBERTa finance-specific models. Results are presented on both the entire dataset and on its robust subset with 100% agreement. In terms of almost all of the

evaluation criteria, FinSentiment clearly outperforms all the existing methods. LSTM with GLoVe approach uses static embeddings, and LSTM with GLoVe classifier cannot integrate model knowledge, so it performs the worst. The performance of LSTM with GLoVe is close to the remaining models in terms of accuracy, but its F1 score is lower. This is because LSTM with GLoVe embeddings performs reasonably better across the neutral class. On the other hand, in terms of all evaluation criteria, LSTM with ELMo embeddings enhances the performance of LSTM with static GLoVe embeddings. LSTM with ELMo embeddings again has a low mean F1 score mainly because of worse performance across sparsely distributed labels. However, its performance is relatively better than LSTM with GLoVe embeddings compared with BERT, XLNet, and RoBERTa in terms of both accuracy and F1 score. According to these results, the results from contextualized word embeddings are closer to ML-based approaches for datasets of this number of samples than the static embeddings. BERT, XLNet, and RoBERTa enhance the performance remarkably on all evaluation metrics, as well as the models do not perform significantly better in certain classes than the remaining classes. According to this result, LM pretraining is effective in prediction performance enhancement for these general models. Among them, RoBERTa slightly outperforms the remaining two ML-based models BERT and XLNet. We would expect all of these larger general models (BERT, XLNet, and RoBERTa) to overfit on this smaller Financial PhraseBank dataset. However, these models can solve this smaller data issue by incorporating efficient training strategies and LM pretraining. Lastly, finance domain-specific FinSentiment clearly outperforms all the existing methods across all metrics, where we report the best of Fin-BERT, Fin-XLNet, and Fin-RoBERTa for the corresponding FinSentiment values. Such outperformance of FinSentiment shows the effectiveness of finance-domain specific training on the sentiment classification task.

In Table 3, we found FinSentiment to outperform all the remaining approaches across the whole set of considered metrics in Financial PhraseBank dataset. We also evaluate the performance of these approaches in Financial PhraseBank dataset across various number of labeled training set samples,

including configurations for 100, 250, 500, 1000, and 3000 samples. We found that training dataset of 100 samples is not enough for all models. Nevertheless, once we start using 250 samples in the training set, FinSentiment begins differentiating between labels efficiently. The performance of all approaches increases by increasing the number of samples in the training dataset. However, FinSentiment performs better with 250 samples than LSTM classifiers perform by using all samples in the training dataset. These results indicate the importance of LM pretraining. We also show the detailed breakdown of FinSentiment as in Table 4 where Fin-Llama outperforms in terms of accuracy whereas Fin-BERT outperforms in terms of loss and F1 metrics when we use all data. The results are similar when we only focus on the dataset with 100% agreement.

Table 5 presents our results for FiQA financial sentiment dataset, where LSTM column denotes the best performance of LSTM with GLoVe and LSTM with ELMo embeddings and FinSentiment column reports the best of Fin-BERT, Fin-XLNet, Fin-RoBERTa, Fin-GPT, Fin-Llama, and Fin-T5 results as stated previously. FinSentiment clearly outperforms all the existing models for both MSE and R^2 . It even outperforms the state-of-the-art models introduced in S. Yang et al. (2018) and Piao and Breslin (2018). All general models without finance-specific training again outperform LSTM models with different embeddings, similar to Financial PhraseBank dataset. We use the reported results for state-of-the-art papers (S. Yang et al. 2018; Piao and Breslin 2018), on which the used test set is the official FiQA Task 1 test set. As we cannot access the full test set used in these papers, 10-fold cross-validation results are reported by us. From this perspective, our models are at a disadvantage as we should keep a subset of the training set as a test set, whereas the state-of-the-art papers utilize the whole training set. LSTM is even outperformed by non-domain-specific BERT, XLNet, and RoBERTa on both metrics, which also shows the importance of language model pretraining. Similar to Table 3 above for classification task, FinSentiment outperforms their nonfinance specific counterparts, which shows the domain-specific training on regression tasks.

TABLE 3 | The best results are shown in bold for the corresponding evaluation metric in Financial PhraseBank dataset.

Model	All data			Data with 100% agreement		
	Loss	Accuracy	F1	Loss	Accuracy	F1
LSTM + GLoVe	0.802	0.714	0.641	0.572	0.817	0.745
LSTM + ELMo	0.741	0.756	0.706	0.504	0.843	0.777
BERT	0.391	0.827	0.816	0.122	0.932	0.943
XLNet	0.392	0.822	0.827	0.152	0.933	0.944
RoBERTa	0.372	0.835	0.834	0.133	0.948	0.966
GPT	0.371	0.851	0.836	0.111	0.955	0.959
Llama	0.350	0.855	0.847	0.102	0.965	0.971
T5	0.362	0.835	0.839	0.130	0.969	0.960
FinSentiment	0.332	0.882	0.864	0.093	0.991	0.988

Note: We report 10-fold cross-validation results.

TABLE 4 | The best results are shown in bold for the corresponding evaluation metric in Financial PhraseBank dataset for each FinSentiment component.

Model	All data			Data with 100% agreement		
	Loss	Accuracy	F1	Loss	Accuracy	F1
Fin-BERT	0.332	0.862	0.864	0.093	0.975	0.988
Fin-XLNet	0.335	0.869	0.851	0.115	0.968	0.965
Fin-RoBERTa	0.391	0.875	0.848	0.123	0.972	0.960
Fin-GPT	0.373	0.874	0.847	0.129	0.975	0.967
Fin-Llama	0.340	0.882	0.855	0.109	0.991	0.980
Fin-T5	0.349	0.870	0.839	0.130	0.964	0.974

Note: We report 10-fold cross-validation results.

TABLE 5 | The best results are shown in bold for the corresponding evaluation metric.

Metric	LSTM	S. Yang et al. (2018)	Piao and Breslin (2018)	BERT	XLNet	RoBERTa	FinSentiment
MSE	0.13	0.08	0.09	0.07	0.07	0.07	0.06
R^2	0.35	0.40	0.41	0.52	0.51	0.55	0.59

Note: The results reported by S. Yang et al. (2018) and Piao and Breslin (2018) are over the official test. Such an original dataset is not available to us, so we report MSE and R^2 results with 10-fold cross-validation.

TABLE 6 | The performance of general (BERT, XLNet, and RoBERTa) and finance-specific models (Fin-BERT, Fin-XLNet, Fin-RoBERTa) in terms of accuracy on three financial sentiment analysis tasks.

Dataset		BERT	Fin-BERT	XLNet	Fin-XLNet	RoBERTa
PhraseBank	0.827	0.862	0.822	0.869	0.835	0.875
FiQA	0.653	0.844	0.662	0.846	0.692	0.882
AnalystTone	0.840	0.887	0.845	0.901	0.843	0.897

Note: The best result for each dataset is shown in bold.

TABLE 7 | The performance of general (GPT, Llama, and T5) and finance-specific models (Fin-GPT, Fin-Llama, and Fin-T5) in terms of accuracy on three financial sentiment analysis tasks.

Dataset	GPT	Fin-GPT	Llama	Fin-Llama	T5	Fin-T5
PhraseBank	0.851	0.874	0.855	0.882	0.841	0.870
FiQA	0.712	0.821	0.731	0.875	0.701	0.818
AnalystTone	0.849	0.895	0.842	0.898	0.835	0.871

Note: The best result for each dataset is shown in bold.

Table 6 analyzes the performance of the contributing finance-specific models in FinSentiment (Fin-BERT, Fin-XLNet, and Fin-RoBERTa) with respect to their general counterparts in detail. In general, finance-specific models, especially Fin-RoBERTa, substantially outperform the general counterparts of the corresponding finance-specific models in terms of accuracy on all three financial sentiment analysis tasks; PhraseBank, FiQA, AnalystTone. The accuracies of all approaches appear to be higher in AnalystTone dataset. Similarly, Table 7 repeats the same analysis for Fin-GPT, Fin-Llama, and Fin-T5 with respect to their general counterparts. Similar to the result in Table 6, finance-specific models, especially Fin-Llama, substantially

outperform the general counterparts of the corresponding finance-specific models in terms of accuracy on the same 3 financial sentiment analysis tasks. Overall, the differences between finance-specific models and their counterparts appear to be quite remarkable for FiQA dataset. We observe similar results in terms of F1 score as well. These results again show the importance of domain-specific training across all three different datasets and tasks.

Table 8 analyzes the performance of the contributing finance-specific models in FinSentiment (Fin-BERT, Fin-XLNet, and Fin-RoBERTa) with respect to their general counterparts in

TABLE 8 | Accuracy results of financially fine-tuned and general models.

Train set	Test set	Fine-tuning	Fin-BERT/BERT	Fin-RoBERTa/RoBERTa	Fin-XLNet/XLNet
FiQAPost	FiQAPost*	Financial	0.502	0.684	0.635
		General	0.502	0.644	0.610
FiQAPost	allAgreeFPD	Financial	0.542	0.720	0.637
		General	0.392	0.723	0.621
FiQAHeadline	FiQAHeadline*	Financial	0.613	0.702	0.557
		General	0.664	0.552	0.536
FiQAHeadline	allAgreeFPD	Financial	0.650	0.808	0.639
		General	0.539	0.777	0.614
allAgreeFPD	FiQAPost	Financial	0.432	0.545	0.484
		General	0.400	0.543	0.460
allAgreeFPD	allAgreeFPD*	Financial	0.985	0.991	0.978
		General	0.972	0.988	0.972
allAgreeFPD	50AgreeFPD	Financial	0.867	0.875	0.872
		General	0.863	0.878	0.873

Note: The dataset is split with 0.3 rate across the scenarios with *.

detail in terms of accuracy where the train and tests are possibly from different cross datasets. In the table, *General* represents the already available general models, and *Financial* represents the finance-specific models that are fine-tuned by using the 10-K corpus. Among the datasets in the table, *allAgreeFPD* denotes the portion of Financial PhraseBank dataset where annotators fully agree on their labeling, whereas *50AgreeFPD* denotes the portion of the same dataset with between 50% and 65% agreement levels. Similarly, *FiQAHeadline* dataset contains only news headlines in FiQA datasets, whereas *FiQAPost* dataset includes the subset of FiQA dataset just with tweets. Even when we train the models by a dataset and test their performance on a different dataset, finance-specific models outperform their general counterparts. The improvement percentage differs depending on the model type and the datasets. For example, when the training dataset is *allAgreeFPD* from Financial Phrase Databank (FPD), unsurprisingly, *allAgreeFPD* perform better when they are tested with dataset again from FPD. Likewise, we have found *FiQAHeadline* and *FiQAPost* to outperform when they are tested with a test set from FiQA in comparison to the test set from FPD. On the contrary, several cases with training set as *FiQAHeadline* or *FiQAPost* show better performance when testing with a set from FPD, even when the test set is extremely different than the training dataset. According to these accuracy results, subject to the dataset, RoBERTa typically outperforms the remaining XLNet and BERT models. General outperformance of RoBERTa is also observed in terms of F1 score across different training and test dataset combinations.

5.2 | The Impact of Additional Pretraining Type

We evaluate the impact of additional pretraining type on the classifier's performance as seen in Table 9. Table 9 shows the

TABLE 9 | The best result for each metric is shown in bold.

Model	Loss	Accuracy	F1
Vanilla BERT	0.39	0.83	0.82
Fin-BERT-task	0.35	0.86	0.87
<u>Fin-BERT-domain</u>	<u>0.34</u>	0.87	<u>0.86</u>

Note: Reported results are obtained with 10-fold cross-validation.

results only for BERT variants. However, results are similar for XLNet and RoBERTa as well. We evaluate three models: (1) Vanilla BERT: Plain BERT model without any additional pretraining, (2) Fin-BERT-task: Fin-BERT model with additional pretraining over the classification train set, and (3) Fin-BERT-domain: Fin-BERT model with additional pretraining over domain corpus, 10Ks, and 10Qs. Table 9 shows the results, where we have evaluated the models with respect to macro average F1 score, accuracy, and loss over the test set. The best result for each metric is shown in bold in the table. The classifier which has been additionally pretrained on the financial domain corpus outperforms the remaining models, even though their relative performance difference is not significant. This result can be mainly explained by the fact that the task set's distribution may be different than the distribution of the corpus.

5.3 | Catastrophic Forgetting Prevention

We assess the performance of our approaches against catastrophic forgetting under five different settings: (1) no adjustment (None) where all layers have the same stable learning rate, (2) only with slanted triangular learning rate (STLR), (3) slanted triangular learning rate and gradual unfreezing (STLR+GU), (4) slanted triangular learning rate and discriminative fine-tuning

TABLE 10 | Catastrophic forgetting results on BERT and RoBERTa models.

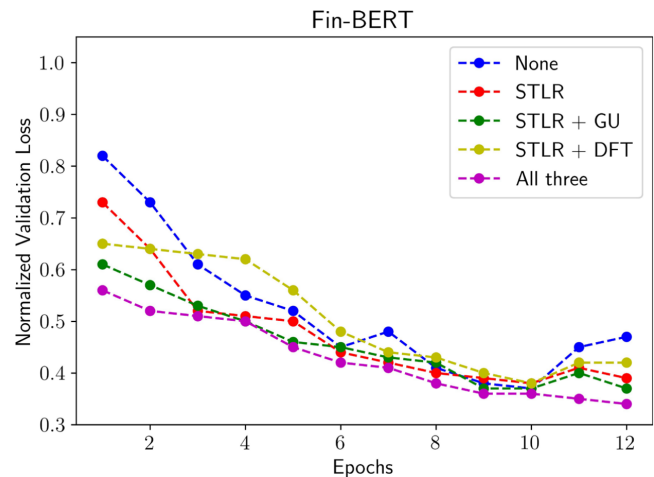
Strategy	Fin-BERT			Fin-RoBERTa		
	Loss	Accuracy	F1	Loss	Accuracy	F1
None	0.474	0.831	0.833	0.455	0.822	0.842
STLR	0.392	0.812	0.813	0.387	0.819	0.831
STLR + GU	0.377	0.858	0.875	0.356	0.891	0.875
STLR + DFT	0.425	0.802	0.799	0.396	0.844	0.807
All three	0.341	0.862	0.856	0.332	0.875	0.864

Note: The best result for each column is shown in bold. The reported results are obtained with 10-fold cross-validation. Abbreviations: DFT, discriminative fine-tuning; GU, gradual unfreezing; STLR, slanted triangular learning rates.

(STLR + DFT), and (5) all three techniques (All three at the same time).

Among these strategies, gradual unfreezing is the strategy in which model layers are unfrozen gradually, and for the discriminative fine-tuning, the applied learning rate is increased from the first to the last layers. Hence, the logic behind these two strategies is alike, that is, for those two techniques, we would expect resembling behavior in the performance of the model. The performance of those five settings is reported in terms of accuracy, loss on test set, and F1 score for both Fin-BERT and Fin-RoBERTa as seen in Table 10. We do not report the results for Fin-XLNet, but its results are similar to BERT. For Fin-BERT, we observe the best performance with regard to accuracy and test loss when all three strategies are applied. The main motivation behind discriminative fine-tuning and gradual unfreezing is the same: Lower-level attributes must be fine-tuned less than higher-level attributes, as the knowledge extracted by language modeling exists largely in the lower levels. As seen in Table 10, using solely STLR performs better than using STLR + DFT, so gradual unfreezing is the most principal strategy in our case. For Fin-RoBERTa, using slanted triangular learning rate and gradual unfreezing to prevent catastrophic forgetting performs better than using all three strategies at the same time, even though we have reported that for the performance of the case, we use all three strategies at the same time in previous sections. Therefore, RoBERTa results in previous sections can be further enhanced by adopting better strategies to prevent catastrophic forgetting.

An unanticipated jump in validation loss after a number of epochs can be one of the observations for the existence of catastrophic forgetting. If no strategy for catastrophic forgetting is applied, as we train the model, it rapidly begins overfitting. According to Figure 1, we observe such overfitting when none of the proposed strategies have been applied. In this case, after achieving the lowest loss on the initial epoch, the model begins overfitting and the performance deteriorates. In other extreme cases, when all the three techniques are carried out, the model will have much greater stability. The remaining strategy mixtures will take place between these two extreme scenarios. Figure 1 shows the validation loss with respect to epochs only for Fin-BERT, but plots are similar for Fin-RoBERTa and Fin-XLNet as well. For the overall process, as expected, STLR + GU, STLR + DFT, and all three strategies applied models behave similarly, that is, their loss values are extremely close and their validation loss change plots are nearly parallel. None and only

**FIGURE 1** | Normalized validation loss graph for Fin-BERT model across various strategies to prevent catastrophic forgetting.

STLR strategies reach the best performance in the initial epochs, and after that, they overfit. However, the remaining strategies perform better and better by increasing epochs.

5.4 | Observing the Impact of Layers

Fin-BERT and Fin-RoBERTa base models have 12 encoder layers, while the Fin-XLNet base model has 12 transformer layers. The last layers in these models do not always capture the most appropriate knowledge on the classification task as part of LM training. So, we have applied sentiment analysis to each layer independently in order to see the contribution of each layer in the classification process and results. In our analysis, the classification layer is placed after the respective embedding's [CLS] tokens in BERT.

In Table 11, we show classification results from different layers over Fin-BERT and Fin-XLNet. Results for Fin-RoBERTa are similar to Fin-BERT. There is a sinus wave-like pattern for the results by increasing layers for Fin-BERT. The best performance is obtained in the Layer 7 with 0.921 accuracy which is 0.111 greater than the lowest accuracy score among the layers, which belongs to Layer 1. However, the performance discrepancy of the last layers is not huge, as it can be observed in Table 11. For Fin-XLNet, there is also a sinus wave-like pattern for the results

by increasing layers with zig-zags on the latest layers. Among the transformer layers in Fin-XLNet, Layer 6 has the best performance results, with the performance variation being in a small interval for the layers between 4 and 8. For all these finance-specific models, the highest scores are observed between the Layers 4 and 8. Additionally, in this interval of layers, the performance does not make a significant difference. According to the table, almost middle layers, 6th and 7th layers, exhibit the best model performance with regard to all considered metrics for Fin-BERT and Fin-XLNet, respectively. Such result may be due to two factors: (1) The trained model becomes larger, so remarkably capable when the higher layers are utilized for classification task. (2) The low layers in the model represent deeper semantic knowledge, so those layers cannot easily fine-tune such knowledge for classification task.

5.5 | The Impact of Training the Model Over a Subset of Layers

Fine-tuning the whole set of models demands remarkable computational power and running time, even for smaller datasets since all general and finance-specific models are enormous. Across several scenarios, a bit lower performance can be preferred if we can achieve such lower performance by fine-tuning a smaller subset of the whole set of parameters. Such modification in fine-tuning may make our FinSentiment models more appropriate to use, principally for larger training datasets. We fine-tune solely the final k many encoder layers as part of our experiments. Fine-tuning solely the classification layer does not perform as well as fine-tuning the remaining layers. Though, clever fine-tuning of only the final layer performs better than most of the state-of-the-art machine learning methods. We observe the almost same performance after Layer 9, which performance is surpassed via fine-tuning the complete

model. We find that a longer training of the complete model is not necessary in order to fully use FinSentiment models. In this case, a trade-off exists between a shorter running time for training and a tiny performance drop in the model.

5.6 | The Impact of Cross-Validation

We generally used 10-fold cross-validation in our experiments. Another common cross-validation in machine learning literature is fivefold cross-validation. In Table 12, we repeat the same experiment as in Table 3 with the only difference being the cross-validation type. Results are presented on both the all dataset and on its robust subset with 100% agreement. FinSentiment again outperforms the rest of the baselines. Scores are also similar to Table 3 where we use 10-fold cross-validation. In general, GPT and Llama still do not outperform comparably older models, mainly due to all of these larger general models overfitting more on this smaller Financial PhraseBank dataset. Such outperformance of FinSentiment shows the effectiveness of finance-domain specific training on sentiment classification tasks.

5.7 | The Performance When Models Are Combined via Bagging

To test the robustness of financially finetuned models, we also report the performance when we apply bagging (Bootstrap Aggregation) (Breiman 1996) and Simple Averaging in our analysis. The results are shown in Table 13. In all train and test cases, bagging outperforms simple averaging. When compared with single finetuned models, bagging performs slightly better in most cases.

TABLE 11 | Accuracy, F1 score, and loss results for each encoder/transformer layer of Fin-BERT and Fin-XLNet models, respectively, when used for classification.

Layer	Fin-BERT			Fin-XLNet		
	Accuracy	F1	Loss	Accuracy	F1	Loss
Layer 1	0.810	0.667	0.525	0.809	0.735	0.488
Layer 2	0.848	0.786	0.397	0.862	0.837	0.374
Layer 3	0.880	0.849	0.334	0.867	0.847	0.374
Layer 4	0.876	0.842	0.341	0.901	0.887	0.281
Layer 5	0.897	0.867	0.303	0.913	0.903	0.273
Layer 6	0.918	0.896	0.256	0.925	0.912	0.250
Layer 7	0.921	0.900	0.250	0.924	0.909	0.262
Layer 8	0.919	0.896	0.286	0.912	0.894	0.259
Layer 9	0.901	0.903	0.307	0.921	0.909	0.283
Layer 10	0.885	0.895	0.315	0.872	0.841	0.336
Layer 11	0.868	0.883	0.324	0.787	0.578	0.470
Layer 12	0.862	0.864	0.334	0.869	0.838	0.374

TABLE 12 | The best results are shown in bold for the corresponding evaluation metric in Financial PhraseBank dataset.

Model	All data			Data with 100% agreement		
	Loss	Accuracy	F1	Loss	Accuracy	F1
LSTM + GLoVe	0.861	0.721	0.640	0.575	0.801	0.741
LSTM + ELMo	0.793	0.745	0.702	0.509	0.820	0.771
BERT	0.403	0.821	0.805	0.124	0.924	0.942
XLNet	0.403	0.818	0.819	0.155	0.932	0.924
RoBERTa	0.400	0.831	0.825	0.132	0.948	0.969
GPT	0.376	0.862	0.838	0.119	0.950	0.969
Llama	0.355	0.850	0.841	0.103	0.968	0.972
T5	0.364	0.832	0.829	0.144	0.963	0.950
FinSentiment	0.344	0.865	0.860	0.095	0.988	0.980

Note: We report fivefold cross-validation results.

TABLE 13 | Accuracy results of performing bagging and averaging of six financially finetuned models.

Train/test set	Bagging	Averaging
FiQAPost/FiQAPost*	0.642	0.611
FiQAPost/allAgreeFPD	0.653	0.728
FiQAHeadline/FiQAHeadline*	0.683	0.722
FiQAHeadline/allAgreeFPD	0.751	0.821
allAgreeFPD/FiQAPost	0.532	0.565
allAgreeFPD/allAgreeFPD*	0.982	0.993
<u>allAgreeFPD/50AgreeFPD</u>	<u>0.858</u>	<u>0.874</u>

Note: The dataset is split with 0.3 rate across the scenarios with *.

5.8 | Analysis of Predictions

We examine the forecasts of both general and finance-specific models by using Financial PhraseBank dataset's 100% annotator agreement part as the test dataset. Such dataset consists of 2264 sentences for which 303, 570, 1391 of them are labeled as negative, positive, and neutral, respectively. Both general and finance-specific models exhibit a very high accuracy over the Financial PhraseBank dataset's 100% annotator agreement subpart. So, examining the scenarios where models fail to forecast the true label will be quite interesting. In this section, we introduce a number of sentences where models make incorrect predictions.

For Fin-BERT model, half of the wrongly predicted sentences are originally neutral and among the rest of the sentences; there is an approximately equal distribution between positive and negative sentences. While negative sentences are mostly predicted as positive (80%), positive sentences are predicted neutral with 88%, and neutral sentences are predicted positive with 91%. In other words, Fin-BERT model is inclined to entangle positive with neutral when it fails to predict correctly.

Moreover, for Fin-RoBERTa model, 42% of the incorrect predictions have the true label positive. The second place for the model's unsuccessful prediction belongs to negative sentences with 33%. In addition, negative sentences are perceived mostly as neutral (96%), as well as positive sentences are mistaken as neutral with 92%. Almost all neutral sentences among the incorrect predictions are predicted as positive (98%). Our Fin-RoBERTa model might confound negative and positive sentences as neutral, plus; it may predict neutral sentences with positive labels. Lastly, for our Fin-XLNet model, incorrect predictions are, in general, originally negative sentences, that is, 47% of the incorrect predictions are originally negative. Neutral and positive sentences are predicted as incorrect almost equally likely, 27% and 24% respectively. Also, negative sentences are predicted positive with 78%. Furthermore, nearly the whole set of the neutral sentences is predicted positive and vice versa. In conclusion, Fin-XLNet tends to predict positively.

Tables 14–16 show a number of examples that are incorrect predictions of the sentences whose true labels are negative, neutral, and positive, respectively, as annotated by the experts in finance. As mentioned above, in terms of sentences whose true label is negative, Fin-XLNet tends to predict negative sentences incorrectly as positive. When Fin-RoBERTa fails to predict negative sentences, it is inclined to predict them as neutral in comparison to the general RoBERTa model which is not financially fine-tuned and might predict negative sentences as positive. In other words, Fin-RoBERTa model stays closer to negative in the sentiment range, when testing with the negative sentence prediction. In terms of sentences which true label is neutral, financially fine-tuned Fin-XLNet, Fin-RoBERTa, and Fin-BERT models mostly predict neutral sentences as positive in their unsuccessful predictions. Lastly, in terms of sentences which true label is positive, all financially tuned models mostly predict neutral sentences as positive in their unsuccessful predictions.

We would expect this behavior for these models, since on the sentiment interval, neutral sentiment is closer to positive sentiment in comparison to negative.

TABLE 14 | The sample of sentences with wrong predictions whose true label is negative.

Text	Fin-BERT/BERT	Fin-RoBERTa/RoBERTa	Fin-XLNet/XLNet
Profit before taxes amounted to EUR 56.5 mn, down from EUR 232.9 mn a year ago.	Positive/neutral	Neutral/neutral	Positive/positive
Salonen added that data shows producers' pulp inventories in North America are declining.	Neutral/neutral	Neutral/neutral	neutral/neutral

TABLE 15 | The sample of sentences with wrong predictions whose true label is neutral.

Text	Fin-BERT/BERT	Fin-RoBERTa/RoBERTa	Fin-XLNet/XLNet
Ahlstrom's share is quoted on the NASDAQ OMX Helsinki.	Positive/positive	Positive/positive	Positive/positive
Kemira shares closed at (x20ac) 16.66 (\$22US.71).	Positive/negative	Positive/negative	Positive/positive

TABLE 16 | The sample of sentences with wrong predictions whose true label is positive.

Text	Fin-BERT/BERT	Fin-RoBERTa/RoBERTa	Fin-XLNet/XLNet
The agreement strengthens our long-term partnership with Nokia Siemens Networks.	Neutral/neutral	Neutral/neutral	Neutral/neutral
Production capacity will increase from 36,000 to 85,000 t per year and the raw material will continue to be recycled paper and board.	Neutral/neutral	Neutral/neutral	Neutral/neutral

6 | Conclusion

In this research, we have come up with FinSentiment model for financial tasks, which incorporates enhanced versions of a number of recently-proposed neural LMs, such as BERT, XLNet, RoBERTa. The corresponding finance-specific models in FinSentiment are called Fin-BERT, Fin-XLNet, and Fin-RoBERTa, respectively. In general, FinSentiment models better perform across sentiment analysis tasks in financial domain by training these models on financial domain corpora. To our best knowledge, this is the first study to investigate the performance of finetuning commonly used pretrained neural LMs in financial domain and tasks. Additionally, it is one of the limited comprehensive studies that focus on additional pretraining over domain-specific corpora. We have evaluated the performance of FinSentiment models over three different sentiment datasets. On all of the used datasets, we have significantly outperformed the state-of-the-art methods. For instance, for sentiment classification tasks, FinSentiment has outperformed the existing generally trained versions and baselines by more than 10% in terms of accuracy and F1 scores.

When compared with the baseline and nonfinance specific models, our results indicate the success of pretrained LMs for a subsequent task such as financial sentiment analysis by using only a tiny labeled dataset. For instance, one of our datasets, Financial PhraseBank, has almost 3000 training samples. However, our FinSentiment models have surpassed the baseline approaches even when we trained with as few as 500 samples. This result is

important since deep learning approaches for natural language processing tasks have, in general, considered to be extremely data-eager. According to the detailed experiments we have carried out with FinSentiment, we conclude that additional pre-training of LMs on a finance-specific corpora has remarkably enhanced the performance. We have found RoBERTa to mostly outperform the remaining ones, and fine-tuning it with the unsupervised financial text data generally improves its performance. We have also investigated the impact of additional pre-training and a number of different training styles. Catastrophic forgetting can be prevented more effectively by using learning rate styles which focus on finetuning the initial layers more strongly than the next layers for all of the considered models.

As a future work, analysis of financial sentiment cannot be an aim alone, and its analysis can be practical once it supports the decisions on financial tasks. As a further study, the proposed finance-specific models and mechanisms can be improved for classification purposes in terms of aspect levels, and regression purposes. Over and above, the model catalog can be expanded. One possible extension will be to use the trained finance-specific models with stock market price datasets over financial tweets and news to predict both the directionality of the returns as well as predicting stock return volatility. Finance-specific FinSentiment models are capable of inferring explicit sentiments. However, modeling the implicit knowledge in the text which is hidden even to domain experts could be a difficult problem. Another future work might to be use the trained finance-specific model over financial text mining tasks

different than financial sentiment analysis tasks, such as question answering and named entity recognition in the financial domain.

Author Contributions

Zehra Erva Ergun contributed to the study design, implemented the methods, analyzed the data, interpreted the results, and prepared the figures. Emre Sefer contributed to the study design, interpreted the results, and finalized the manuscript. The authors read and approved the final manuscript.

Ethics Statement

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Our source code has been deposited at <https://github.com/seferlab/fin-sentiment>.

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